

Business Forecasting System

Multi-model, backtested forecasting for new patients, clinical encounters, collections, and contact lens sales — with every model competing on real held-out accuracy, not assumption.

6

CANDIDATE MODELS

4

BUSINESS METRICS

33-95

MONTHS OF HISTORY

100%

BACKTESTED
SELECTION

What feeds the system

Four business numbers, one value per month, pulled directly from the practice management database — summarized only, never row-level patient records.

METRIC	SOURCE TABLE	DATE COLUMN	VALUE
New Patients	Customers	CREATE_DATE	Count of new records / month
Encounters	PATIENT_ENCOUNTERS	ENCOUNTER_DATE	Count of visits / month
Collections	PMS_PAYMENTS	PAYMENT_RECEIVED_DATE	Sum of payments / month
Contact Lenses	PMS_OPTICAL_ORDERS + details	ORDER_DATE (CREATE_DATE fallback)	Sum of lens units / month

DATA CLEANING — BEFORE A SINGLE MODEL SEES IT

Anomaly detection

A statistical rule (robust z-score) automatically flags whole months that look like data artifacts — e.g. a one-time system migration dumping thousands of fake records into a single month — and excludes them, without hardcoding any specific date.

Transaction-level filtering

For collections specifically, individual fake/test payments (like a \$1,111,111.00 test entry) are detected and removed *before* they're summed into a monthly total — so one bad row can't distort an entire month.

Real example caught this way: a database migration artifact inflated one month's patient count to 4,749 against a normal baseline of 40-85/month — automatically excluded, with the reason logged.

Also caught: ~48% of contact lens order records had a placeholder date (1900-01-01) from a legacy data path — recovered using a fallback date column instead of silently losing nearly half the history.

Six models compete — nothing is assumed

Every metric is forecast by all six candidates below. Whichever one is demonstrably most accurate on real historical held-out months wins — separately, per metric.

MODEL	TYPE	WHAT IT USES
Seasonal Naive	Baseline	Same month, one year ago
ETS	Statistical	Trend + seasonality smoothing
SARIMA	Statistical	Autoregressive time-series structure
Prophet	Statistical	Seasonality + holiday effects
XGBoost	Machine learning	Calendar + lags + other metrics as features
SARIMAX	Statistical + ML hybrid	SARIMA structure + the same extra features

THE TRAINING / EVALUATION PIPELINE



TWO SAFETY CHECKS ON TOP OF RAW ACCURACY

🚩 Flat-forecast detection

If a model's "prediction" is really just a flat, unchanging number every month — a sign it gave up on finding real structure — it is disqualified, even if its raw error score looked best. Checked both during backtesting *and* again on the final production forecast, since refitting on full history can behave differently than on backtest folds.

📊 Plain-language accuracy

Results are shown as an intuitive 0-100% accuracy score (100% minus the error rate), not a raw statistical error metric — usable by a business owner, not just a data analyst.

What columns actually feed the models

Every model starts from the same base — one row per month, a date and a value. The four statistical baselines never see anything more. XGBoost and SARIMAX get additional engineered columns, kept only where they measurably improve backtested accuracy.

CALENDAR

Month/quarter as sine-cosine pairs (so December and January read as "close," not 11 apart), US holiday count per month, January/December/summer/Q4 flags.

SELF-HISTORY

The metric's own value 1, 2, 3, 6, and 12 months ago; rolling 3/6/12-month averages; month-over-month and year-over-year change.

CROSS-METRIC SIGNALS

The *other* three metrics' values from 1 and 3 months ago — never the same month, since that would use information not yet available at forecast time.

Why not use every possible feature? With only 33-95 months of data, more engineered columns than that risks the model memorizing noise instead of real pattern. This was proven empirically this cycle — SARIMAX (SARIMA plus these same extra features) performed *worse* than plain SARIMA on the shorter series, a textbook overfitting signature.

HOW THE FINAL HEADLINE NUMBER IS BUILT

- 1 **Actual, already-happened numbers** for the completed months so far this year.
- 2 **This month (in progress)** — estimated from a day-by-day pacing curve learned from history (e.g. "by day 15, a typical month is usually 45% complete"), not from the whole-month statistical model, since a partial month isn't a real data point yet.
- 3 **Future months** — the winning model's forecast for months that haven't started.

Add all three together for "by end of year." The same forecast independently produces "next month" and a rolling "next 12 months" figure.

Real, backtested accuracy — per metric

Bars show each model's held-out accuracy (higher is better). The gold star marks the model actually used in production for that metric.



New Patients

Winner: ETS (Holt-Winters) · 76.6% accuracy

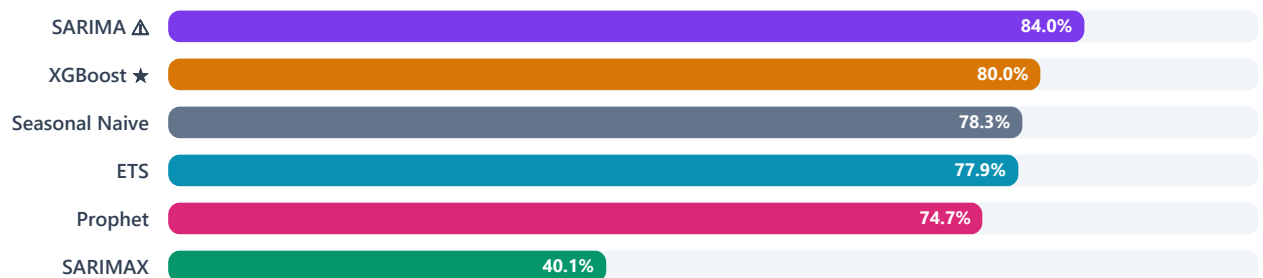


Δ SARIMA scored highest on raw error but was disqualified — its forecast was flat/constant, not real structure.



Clinical Encounters

Winner: XGBoost (multi-feature) · 80.0% accuracy



Δ Same disqualification pattern as patients — SARIMA's raw score looked best but was flat.



Collections (\$)

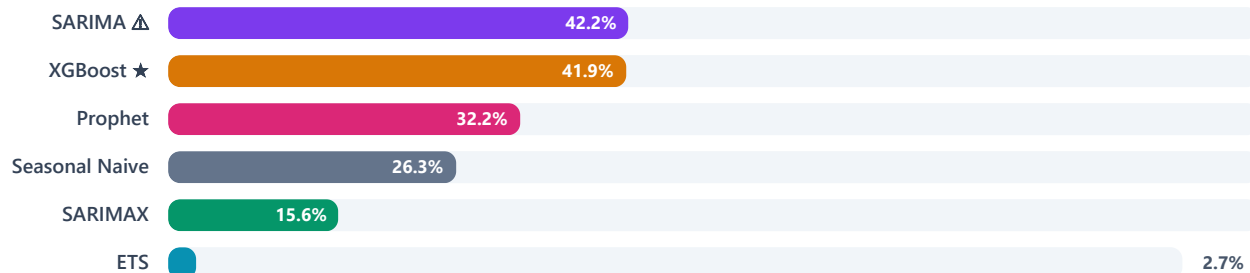
Winner: Seasonal Naive · 89.3% accuracy





Contact Lenses Sold

Winner: XGBoost (multi-feature) · 41.9% accuracy



Honest read, not a spin: ~42% accuracy is the weakest of the four metrics — a real limitation, not hidden. With only 36 months of genuinely clean order data, there simply isn't enough learnable pattern yet. Treat this number as directional, not a commitment. The roadmap's top recommendation to fix this specifically: use contact-lens exam *bookings* as a leading indicator, since they precede the actual order by days to weeks.

INTERPRETING THESE NUMBERS CORRECTLY

Use the range, not just the point

Every forecast ships with a 95% confidence range alongside the single number. A 75-90% accurate model can still be meaningfully wrong in any one given month — plan against the range for anything with real financial or staffing stakes.

Accuracy is metric-specific, not universal

There is no single "best model" — SARIMA, ETS, XGBoost, and Seasonal Naive each win for a different metric here, and the winner can change again as more real history accumulates. The system re-evaluates every time it's re-trained rather than assuming last time's answer still holds.

What you actually see and can do



Per-metric dashboards

This month, next month, by year-end, and rolling 12-month figures, growth vs. last year, and a full monthly trend chart — for each of the 4 metrics.



Compare every model yourself

All 6 models' forecasts are computed and saved, not just the winner — pick any of them, view a side-by-side comparison table, or overlay every model on one chart.



Training pipeline console

Re-run fetch → clean → train → evaluate on demand, watch it happen live, and preview exactly which months get cleaned and why before trusting the result.



Multi-database ready

Switch the entire system to a different database (e.g. QA to production) from a Settings page — no code changes, same table/column assumptions apply everywhere.

IN ONE SENTENCE

The system doesn't guess which model is "smart" — it makes six candidates prove themselves against real historical outcomes every time, picks whichever one actually got closest for that specific metric, and builds the headline number from real year-to-date figures plus that model's forecast for what hasn't happened yet.